Felipe S. **Oliveira**

Software Engineeer

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Experience

Faire Wholesale Inc. — *Remote, Canada*

Backend Software Engineer - *Mar 2022 – Present*

- Led the extraction of the Insider membership domain from a Kotlin monolith into a standalone service, defining service boundaries, establishing independent deployment pipelines, and producing migration patterns adopted as a reference for Faire's broader multi-region architecture initiative.
- Designed and implemented asynchronous shipping-estimate precomputation triggered on cart updates via SQS, enabling in-cart shipping cost display for the first time and establishing an event-driven infrastructure pattern that reduced downstream checkout latency from 1.5 s to under 50 ms.
- Operate and evolve Kotlin-based backend services on AWS, using MySQL and CockroachDB for transactional storage, DynamoDB for workflow-driven data, Snowflake for analytics, and gRPC for service-to-service and frontend communication.
- Collaborate across product, data, and infrastructure teams to design scalable systems, improve platform reliability, and reduce cross-cutting technical debt in support of Faire's Insider membership program.

SiDi (Samsung Institute for the Development of Informatics) — *Campinas, Brazil*

Systems Software Engineer - *May 2019 – Feb 2022*

- Led a major architectural refactor of Samsung's eSE-based credential platform (Universal Credential Manager), redesigning interfaces across trusted application and framework layers to improve testability and extensibility, enabling faster feature integration by partner teams in South Korea.
- Designed cross-layer platform APIs for Samsung's Digital Key solution, building secure communication flows between the Android Framework and the embedded secure element (eSE) through ARM TrustZone and exposing framework capabilities for third-party developers via Samsung Knox.
- Researched and delivered the first proof of concept for HDM Detector, a Samsung Knox platform feature for validating application versions on managed devices and enforcing policy-driven remediation. Contributed across Java framework APIs, native layers, trusted application connectors, and TrustZone components.
- Developed Bootloader Forensics, a bootloader-level platform security feature for validating critical system state, detecting tampering, and securely collecting signed and encrypted forensic data as an extension of Samsung's Verified Boot chain.
- Designed and implemented security and systems software for Samsung Knox across the Android Framework, native C/C++ layers, embedded secure elements, and ARM TrustZone trusted applications.

Samsung — *Campinas, Brazil*

Software Security Analyst - *May 2017 – Apr 2019*

- Performed security assessments across Samsung's mobile, embedded, and web platforms — including Android system services, Tizen applications, backend services, and IoT devices — combining reverse engineering, exploit development, and security design analysis to identify vulnerabilities and drive remediation.
- Built internal tooling to reverse engineer gRPC protocols and assess ML-facing attack surfaces during a comprehensive security evaluation of Samsung's Bixby voice assistant, covering client applications and backend services.
- Identified privilege escalation and unauthorized access vulnerabilities in Android system services across Samsung flagship devices; developed proof-of-concept exploits and produced technical reports guiding remediation by platform engineering teams.

Kryptus – Solutions in Information Security — *Campinas, Brazil*

Software Engineer - *Sept. 2016 – May 2017*

- Contributed to a full-disk encryption platform for ATMs based on VeraCrypt, implementing low-level bootloader and pre-boot functionality in C and x86 assembly with custom key storage in BIOS-adjacent NVRAM and support for remote boot, monitoring, and management via iPXE.
- Also contributed to related Windows driver work in C++ and to the server-side remote management platform built in Python with Django.

Education

State University of Campinas — *Campinas, Brazil*

Hon. B.Sc. in Computer Science - *Jan. 2011 - Dec. 2016*

University of Toronto — *Toronto, Canada*

Visiting Student - *Jan. 2014 - Jan. 2015*

Extracurricular Activities and Projects

Linux Foundation *Online Course*

LFD430 - Developing Linux Device Drivers - *2025*

Completed an online course on Linux device driver development offered by the Linux Foundation. The course covered topics such as kernel architecture, module programming, character and block device drivers, and debugging techniques. Gained hands-on experience in writing and testing device drivers for various hardware components.

Honors & Awards

2016	**Graduated with Distinction**, Unicamp	*Campinas, Brazil*
2014	**Grand Prize Winner**, SPORTSHACK 2014	*Toronto, CA*
2014	**Science Without Borders Scholarship**, Brazilian Government	*Brazil*

Skills

Languages	Kotlin, Python, C, C++, Java, Rust, Go (familiar), SQL, Bash
Infrastructure & Cloud	AWS (ECS, SQS, DynamoDB, IAM), Kubernetes (familiar), Docker, Terraform (familiar), CI/CD Pipelines
Observability & Reliability	OpenTelemetry (familiar), Distributed Tracing, Structured Logging, SLI/SLO Definition
Data & Messaging	MySQL, CockroachDB, DynamoDB, Snowflake, SQS, gRPC, Protocol Buffers
Systems & Security	Linux Internals, ARM TrustZone, Android Framework, Cryptography, Secure Boot, Embedded Systems
Disciplines	System Design, Distributed Systems, Service Decomposition, API Design, Developer Tooling, Performance Optimization
Tools	Git, Gradle, Bazel (familiar), JIRA, Perforce
Spoken Languages	English (fluent), Portuguese (native), Spanish (basic)